

Manual

Material and Inventory Management MPL-MBV 8.2

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1 Material and Inventory Management

Purpose

The material and inventory management function allows for material/batches to be managed as well as classified and grouped in material types. These material types specifically control data collection and processing in MES.

The function package also provides functions to edit and, if necessary, correct data of collected batches at a later point in time.

Implementation Considerations

Use this function package to:

- control system performance relating to batches subject to the material.
- to correct data of recorded batches subsequently

Integration

The material type, which is a fundamental element of order data as well as of component data or the used batches, integrates the function package with a variety of function groups.

Features

- Configuration of material types
 - Editing function of material type master data to control material-specific processing
- Configuration of assignment of materials to material types
 - Editing function to classify material in material types
- Configuration of material buffers
 - Editing function to create master data for material buffers
- Configuration of inventory limits
 - Function to edit inventory limits of material buffers
- Configuration of transport units
 - Editing function to classify materials based on transport units
- Configuration to assign transport units to material types
 - Editing function to assign transport units to the configured material types
- Editing and correction of batches
 - Editing and correction functions for recorded batches in batch data overview
- Display and usage of recorded material movements
 - Goods receipts
 - Consumption

- Manual or automatic assignment of batch numbers
 - Manual or automatic assignment of batches or batch numbers according to configuration
- Deletion of last batch
 - Usage of the output batch generated at last according to configuration
- Consumption recording
 - Recording of batch consumption according to configuration

2 Applications Available in MOC

Application	Documentation	Transaction code	Function authorization
Material type	MOC_MaterialType.pdf	mtyp	mtyp
Assignment of material to material type	MOC_AssignmentMaterialMaterialtype.pdf	asmm	asmm
Material buffer	MOC_MasterDataMaterialBuffer.pdf	mbuf	mbuf
Inventory limits	MOC_InventoryLimits.pdf	invl	invl
Transport units	MOC_TransportUnits.pdf	tu	tu
Assignment of transport unit to material type	MOC_AssignmentTransportUnitMaterialtype.pdf	astrum	astrum
Batch data overview	MOC_BatchOverview.pdf	batov	batov
Material movements	MOC_MaterialMovements.pdf	mmov	mmov

3 Material Types

Overview

	→	
	→	
HYDRA menu	Master data → Material → Material types	
FEDRA menu	Detailed scheduling → Master data → Material types	
Transaction code	mtyp	
Function authorization	mtyp	

Available user fields		
Where	Object type/user field key	Source (type)
Detail view	MATTYP/SYSTEM	Material type (MF-D)
How to configure user fields? Which user field types are available?		

Purpose

Use this function to create or modify material types in the system.

Integration

The system uses the material type and the settings stored with it to control the system behavior of objects that are assigned to this material type.

Selection criteria

Material type

Unique material type key

Type

Category of the material type

Description

Description or name of the material type (plain text)

Field descriptions - General

Material type

Unique material type key

Batch size

Typical batch size for batches of this material type.

Description

Description/name (plain text) of the material type

Type

Category of the material type

You can assign a type to group and classify material types that belong to the same category. Assigning a type, however, does not control the processing.

Unit

Unit used for materials of this type.

Field descriptions – Input batch processing**Inventory management**

E - Yes, when logging input batch off

Here, you manually record the used quantity of a batch or the remaining quantity when you log off the batch. Once the batch is logged off, quantities are reduced and the consumption is uploaded to the higher-level systems.

R - Yes, backflush (retrograde)

Once the batch is logged off and the operation is interrupted or logged off, the consumption values are reported to the higher-level system.

G - Yes, backflush (only with output batch = yield)

Same procedure as with R; but the quantity of the input material is only reduced in a retrograde manner if the output batch produced is a yield quantity (batch class = G).

N - No inventory management. Batches of this material type are not included in the stock overview.

Note: Depending on the configuration, you can enter the remaining quantity or the consumption. In general, you can do so, when changing input batches via the Windows terminal. When logging off the input batch, this quantity is posted to the respective batch. This applies for options E, R, G and N.

A – Anonymous inventory management

Batches of this material type are processed as material without batch reference when it comes to composition.

Log off input batches (when OP is interrupted/logged off)

A – Input batches are automatically logged off when an operation is interrupted or logged off and then get the batch status "processed".

F – Input batches are automatically logged off when an operation is interrupted or logged off and then get the batch status "free".

G – Input batches are logged off when an operation is interrupted or logged off according to the configurations in the basic parameter settings.

N – Input batches are not automatically logged off when an operation is interrupted or logged off.

S – Input batches are automatically logged off when an operation is interrupted or logged off and then get the batch status "blocked".

Tolerance when logging off

Value in %.

When the input batch is logged off, the system checks if the absolute value of the remaining quantity is less than the percentage value of the original quantity specified in this field. If this is the case, the batch is set to batch status *Processed* and identified as consumed.

Maximum value

The maximum value specifies the tolerance when logging off.

If the remaining quantity is less than the specified percentage value but greater than the maximum value, the batch is not set to batch status *Processed*.

Standard quantity of unknown input batches

If a quantity for unknown input batches is specified and an unknown input batch of this material type is recorded, this unknown input batch will be entered with the specified quantity. If no value is specified, the system generates the unknown input batch with the quantity 1,000,000,000.

Options

Can be logged on several times to one machine

You can use this option to log on an input batch to a machine for multiple items of the component list (BOM items) of an operation.

If the option is not set, the system issues an error message if you try to log on an input batch that is already logged on once more.

Example:

The following material components are assigned to an operation:

BOM item 0001	Material 487191
BOM item 0002	Material 923948
BOM item 0003	Material 487191
BOM item 0004	Material 574857
BOM item 0005	Material 487191

If the option is enabled, batch 000007153 with material number 487191 can be logged on to one of the three items 0001, 0003, 0005 or to two or even to all of these items. If the batch is already logged on, the system shows a selection window every time you attempt to log on the batch once more. The window shows the available BOM items of the material. To log on the batch once more,

the user has to select the required item. After confirmation, the batch is logged on to this (additional) item of the BOM. If the user selects a BOM item where the batch is already logged on, the labeling of the button changes and the respective batch can be logged off.



In general, different batches can be logged on several times for the same material (same material and different BOM items), irrespective of how this option is set. If this option is enabled, the system assumes that the option *Can be logged on to several machines at the same time* is also enabled.

This option does not allow to log on unplanned material.



In case of merged batches, you cannot use the option *Can be logged on several times to one machine*.

Can be logged on to several machines at the same time

You can log on the input batch to several workplaces and/or operations at the same time. In this case, it does not make any difference if the input batch that is already logged on to an operation at a machine

- is logged on to another operation of the same machine,
- is logged on to another operation of another machine,
- is logged on to the same operation of another machine.



In case of merged batches, you cannot use the option *Can be logged to several machines at the same time*.

Input batch must be logged on

This option activates an additional validation check when an operation is logged on that requires batch management. For all material components (BOM items) with a material type where this check is enabled, the system checks if batches have already been logged on. If the check has a negative result, the logon of the operation is refused.

If you want all materials of the component list to be logged on as input batches before the OP can be logged on, you can configure this via the material type. To do so, use the option *Log OP on only if all components are logged on as inp. batches* of the tab *Output batch processing*.



The system performs the validation check **ONLY** when you log on the operation.

When an input batch is changed using the function *Input batch change*, no check takes

place.

That means: In case of an input batch that must be logged on (see this option), you can log off this input batch, without logging on a new batch for the respective BOM item.

Hand batch number down

If you log on an input batch with a material type where this option is set when the order starts, the system uses this batch number as batch number for the output batch to be logged on.

At the same time, the system checks if the batch number of the input batch may be handed down when the output batch is logged on, i.e. the system checks if the option is enabled for the material type of the input batch.

This option is processed and [run-through batch processing](#) in combination with the [configured](#).



If this option is set, you also have to set the option "Input batch only valid for exactly 1 output batch".

Input batch only valid for exactly 1 output batch

If this option is enabled for the material type of a component, you must log on at least 1 batch as input batch for each output batch.

Note: This option only controls how the batch change is processed (CA_WL). This option does not ensure that an input batch can be logged on again at a later point in time (when another output batch is active).

This option is processed and [run-through batch processing](#) in combination with the [configured](#).



If you use this option without run-through batch processing, incorrect batch assignments can occur during tracing. In this case, the succeeding input batch is linked to the output batch that must actually be logged off.

Allow entry of unknown input batches

Using this option, the system can automatically generate and log on input batches that the system didn't know beforehand.

If you enable this option and if the input batch is not known in the system when you log on the input batch, then the system creates this unknown batch using the selected BOM item (material). The batch is created with a quantity of 1000000000 in primary quantity unit of the operation and a goods receipt posting is performed.

To identify the correct material type, a configuration/assignment between [material and material type](#) must be available so that the system can uniquely identify the material type using the material number of the component. If a material is not assigned to a material type, the material type *SYSTEM* is automatically assigned.



Requirement: You must activate the recording of unknown batches in the basic settings using the option *MPL → Create unknown batches*.

Use data for output material

This option specifies if the system uses the expiry date of the input batch to calculate the expiry date of the output batch. Note: This option is only available if a logged on input batch has a material type where this option is enabled. Only in this case the generated output batch can use the expiry date of the input batch if configured accordingly (see "Determination of expiry date" in the "General" tab).

Decision on changing the input batch

If this option is enabled, the system displays a warning message when the input batch is changed and before the current input batch is logged off.

Especially when serial numbers are collected, this warning message informs the user that all components with a serial number must be evaluated before logging the input batch off. Only then, the components can be "properly" linked to the resulting output batch (merged batch).

Pass batch attributes on

You can use this option to transfer configured batch attributes of an input batch to the resulting output batches. The configuration can be complemented flexibly using the extended object configuration. The configuration is described in the following document: [Configuration to pass on batch attributes](#)

Pass document links on

You can use this option to transfer document links of an input batch to the resulting output batches. The configuration can be complemented flexibly using the extended object configuration. The configuration is described in the following document: [Configuration to pass on document links](#)

Field descriptions – Output batch processing

Min. storage time

Minimum storage time of the material type

The system uses this value to calculate the availability date when the output batch is generated (by adding it to the manufacturing date/time). As long as this calculated point in time is in the future, a batch remains in "Min. storage time" status and cannot be logged on.

Note: You can use the configuration [Assignment material to material type](#) to override this setting for specific material.

Warning limit

Warning limit of the material type

The system uses this value to calculate the respective warning date when the output batch is generated (by adding it to the manufacturing date/time). You can use the function [Warning report](#), for example, to integrate this value in evaluations.

Note: You can use the configuration [Assignment material to material type](#) to override this setting for specific material.

Expiry limit

Expiry limit of the material type

The system uses this value to calculate the respective expiry date when the output batch is generated (by adding it to the manufacturing date/time). At this point in time, the batch is automatically set to "Expired" status and can no longer be logged on.

Note: You can use the configuration [Assignment material to material type](#) to override this setting for specific material.

Reserved when batch is generated

You can use this option to automatically reserve the output batches produced for subsequent processes. You can make reservations on different levels.

Values:

- - No automatic reservation when batches are generated
- - Reservation for all OPs of the order
- - Reservation for subsequent OPs

With the last two values, the system checks if a reservation is available when an input batch is logged on subsequently.

In coil-based production (MPL-RF) and with so-called cutting machines, the reservation is always made for the subsequent mother operation.

Options

Input batch change mandatory if remaining qty. ≤ 0

If this option is enabled, the currently logged on input batches are highlighted in red that have a remaining quantity, which is less than or equal to 0, when an output batch is changed. These input batches must now be logged off.



This option is only applicable for MPL machines (batch management) with machine type "M". The terminal processes this option locally.



This option is only applicable for the dialog *Output batch change* (CA_WL_MPL). The remaining quantity displayed for the logged on input batch must be equal to or less than zero (≤ 0). The system can only perform this check, if the remaining quantity of the input batch is less than or

equal to 0 (≤ 0) before the posting.

Log OP on only if all components are logged on as inp. batches

If this option is set, the OP using the output material can only be logged on if all materials of the component list are logged on as input batches.

If this option is not set, the OP can be logged on even if not all of the materials are logged on as input batches.

Consumption balance

When you log off an OP, the system opens an additional dialog (V_BLZ) displaying the material components and their consumption quantities in relation to the OP that is currently logged on. In this dialog, you can also log off input batches that are still running. The option only becomes active when the consumption balance of the machine has been activated in the "MPL" tab.

Serial no. requirement (MPL-SNR, as of MPL 7.2)

If the system records serial numbers (ADE-SNR), and if this option is active and the operations require batch management, then the system generates an additional batch per serial number in the MPL module. Batches are only collected per serial number (using command A_TR) if the operations require serial number management = E. In the table "Batch relationships" (mpl_beziehungen), traceability to the current output batch (ID) is established using the serial number recorded.

Generation of HU (MPL-SNR, as of MPL 7.2)

If the option "Serial number requirement" is set, you can use this option to control if the serial numbers recorded are stored in relation to a handling unit (merged batch). The option is currently only intended for customizations and does not have any effect.

Note: The system only checks this option when the OP is logged on and when the output batch is changed. When an input batch is changed using the function *Input batch change*, no check is performed.

Log off/interrupt OP without the last batch

If the option is set, a last output batch is generated for the article of the operation (matching this material type) between the last output batch change and/or OP logon and the OP interruption and/or OP logoff, but this output batch is immediately deleted (batch status = "D"). The batch is not visible in the batch history. For further information on this function, refer to the document ["Interrupting/Logging off OP without the Last Batch"](#).

Delete batch assignment

If this option is enabled, the link between the output batch "deleted last" to the currently running input batches of the OP is deleted. As a result, the deleted batch is not visible in batch tracing.

Automatic assignment of serial numbers

This option enables the automatic assignment of numbers for a new part when merging components listed by serial numbers. If this option is not set, you can assign the serial number manually. The option is only relevant if the option "Superordinate serial number" = N is set for the component.

User field key for output batches

The defined user field key is transferred to the generated output batches.

Behavior when output batch has quantity 0

Use the option *Behavior when output batch has quantity 0* to specify system behavior when output batches are changed and the quantity is 0. You can choose from the following three configuration options.

- | | |
|------------------------|--|
| "empty" | No reaction. Output batches are changed with quantity 0. |
| "Warning" | A warning message is displayed. You can still override the warning message. If you override the warning message, output batches are changed with quantity 0. |
| "Error message" | An error message is displayed. The error message cannot be skipped. The output batches cannot be changed with quantity 0. |

Standard quantity of output batches

If a quantity is entered for output batches and an output batch of this material type is generated, the *quantity* field of the shop floor client is populated with the specified quantity. You can still change this default assignment at any time.

Field descriptions - General

Determination of expiry date

- **Current point in time**
The expiry date is calculated using the current point in time and the duration stored for the material type of the output batch.
- **Shift date**
The expiry date of the generated output batch is identified using the current shift date, the time 00:00 and the expiry limit stored for the material type.
- **Input batch**
The expiry date of the generated output batch is identified using the running input batch where the option "Use data for output material" is set for the material type.
- **Processing based on enhanced object configuration**
An additional configuration in the [Advanced object configuration](#) specifies how the expiry date of the generated output batch is identified:

○ **Use the closest expiry date of all logged on input batches**

If this configuration is active, the expiry and warning time of the input batch with the closest expiry time is used for the output batch. The availability date is calculated using the configuration in the "Output batch processing" tab.

Parameter name	Parameter value
Object type	MATTYP
Object ID 1	Material type of the output batch. The configuration is valid for this material type.
Parameter	DATE_OF_EXPIRY_CALCULATION
Parameter value	MIN_DOE_OF_INPUT_MATERIAL

○ **Use the closest expiry date of selected input batches**

If this configuration is active, the expiry and warning time of the input batch with the closest expiry time is used for the output batch. To identify the closest expiry date, the system only uses input batches of the configured material type. The availability date is calculated using the configuration in the "Output batch processing" tab.

Parameter name	Parameter value
Object type	MATTYP
Object ID 1	Material type of the output batch. The configuration is valid for this material type.
Parameter	DATE_OF_EXPIRY_CALCULATION
Parameter value	MIN_DOE_OF_SELECTED_INPUT_MATERIAL
When the relevant input batches are identified, the system only uses the batches that can be identified via an additional configuration for the material type.	
Parameter name	Parameter value
Object type	MATTYP
Object ID 1	Material type for which the configuration applies
Parameter	DATE_OF_EXPIRY_CALCULATION
Parameter value	INCLUDE_IN_DOE_SELECTION

WIP material (Work in Process)

This option specifies whether a batch of this material type is automatically logged on as input batch for the subsequent operation (according to the operation sequence of a production order) (as [WIP material](#)). Find a description of how to configure this function [here](#).

Options

Transfer to interface (goods movements)

If this option is enabled, the produced output batch is identified as goods receipt or the used up input batch as goods issue. These [material movements](#) must then be uploaded.

Hold back goods receipt until usage decision is made

You can use this option to delay the transfer of goods receipts to the PPS system. In this case, the output batch is transferred only if a usage decision is taken.

Field descriptions - Transport

Generate transport order for output material

If this option is enabled, the system creates a [transport order](#) for a batch when an output batch is produced. The transport starts from the material buffer that includes the output batch. This option overrides the configuration of the respective option in the workplace configuration.

Generate transport order for input material

Use this option to create a [transport order](#) for an article (and relating to a material component) when you plan an operation at a machine using the shop floor scheduling module. Transport starts from the output material buffer of the preceding operation. This option overrides the configuration of the respective option in the workplace configuration.

Field descriptions – User fields

You can customize the system to release user fields for the object type "MATTP".

4 1 Assignment Material - Material Type

Overview

Menu	Master data → Material → Material assignment - Material type
Transaction code	asmm
Function authorization	asmm

Usage

This function is used to assign the appropriate [material type](#) to a material in the system.

Integration

Each material can be assigned to a [material type](#). The assignment is used

- if a goods receipt batch is recorded manually
- if an unknown batch is defined

Requirement

The [material type](#) must be defined in the system.

Selection criteria

The following selection criteria are available in the application:

Material

Only the selected materials are selected.

Material type

Only materials with the selected [material type](#) are selected.

When using multiple selection criteria - if nothing else is specified - the amount of overlap of the selection criteria is displayed.

Field descriptions

Material

Material number that is to be assigned to the [material type](#).

Material type

Assigned material type. This must be defined in the system.

Comment

Additional text regarding the material, comments

Min. storage time

Minimum storage time for the material type

During batch determination, the corresponding availability date is calculated using this value (by adding it to the date of manufacture). Up to that point, a batch remains in "Min. storage time" status and cannot be registered.

Please note: this setting overrides the setting with the same name in the *Material type* configuration.

Warning limit

Warning limit of the material type

During batch determination, the corresponding warning date is calculated using this value (by adding it to the date of manufacture). This can be used for an evaluation, e.g. using the function *Warning report*.

Please note: this setting overrides the setting with the same name in the *Material type* configuration.

Expiry limit

Expiry limit of the material type

During batch determination, the corresponding expiry date is calculated using this value (by adding it to the date of manufacture). At this point in time the batch is automatically set to "Expired" status, so it can no longer be registered.

this setting overrides the setting with the same name in the *Material type* configuration.

Notes

A material can only be assigned to exactly one [material type](#). However, one material type can be assigned several materials. The saving procedure for material is case sensitive. The so-called wildcards "*" and "?" can be used in the material field. The selected material type must exist during creation. Otherwise, the creation will be refused.

If a material is assigned to another material type, the material type is NOT automatically updated

- for existing orders/operations
- for existing component lists
- in the existing batches

5 Material Buffer

HYDRA menu	Master data → Material → Material buffer
FEDRA menu	Detailed scheduling → Master data → Material buffer
Transaction code	mbuf
Function authorization	mbuf

Purpose

Use this function to create or modify material buffers in the system.

Integration

The system uses material buffers to represent both physical and virtual material storage or areas where materials are kept for temporary storage in the production process. Material buffers are identified by unique keys.

Selection criteria

The application provides the following selection criteria:

Material buffer

Enter the [material buffers](#) ID to select the material buffer.

Name

Enter the name of the [material buffers](#) to select the material buffer.

Type

Enter the material buffer type to select the material buffer. Possible values:

- F – production buffer
- H - hierarchical buffers
- W - goods receipt buffers
- C – casting buffers (composition)

Cost center

Enter the cost centers the [material buffers](#) are assigned to in order to select the material buffers.

Company

Enter the companies the [material buffers](#) are assigned to in order to select the material buffers.

Department

Enter the departments the [material buffers](#) are assigned to in order to select the material buffers.

Area

Enter the areas the [material buffers](#) are assigned to in order to select the material buffers.



If several selection criteria are entered and unless otherwise specified, the application shows the data that match all criteria.

Field descriptions - General

Material buffer

ID of the [material buffer](#) that is created or edited.

Type

Material buffer type. Possible values:

- F – production buffer
Material buffers located in the shop floor are referred to as production buffers. Storage locations and material buffers of resources should always be configured as production buffers.
- H – hierarchical buffer
A hierarchical buffer combines several individual buffers. The quantity of individual buffers is accumulated in the hierarchical buffer for evaluations and reports.
- W – goods receipt buffer
Material buffers are referred to as goods receipt buffers if material is supplied externally. They are the source for material used in production.

Production buffers and goods receipt buffers are processed identically. In contrast, a hierarchical buffer is a "virtual" buffer enabling the combination of one or more buffers (also hierarchical buffers), so that evaluations can be made on this basis in the [Stock overview](#) function (menu: material management --> inventory management).

Name

Long text of the [material buffer](#).

Storage location

Refers to the physical storage location of the material buffer (currently not used).

Department

Department the [material buffer](#) belongs to.

Area

Area the [material buffer](#) belongs to.

Cost center

Cost center the [material buffer](#) belongs to.

Company

Company the [material buffer](#) belongs to.

Comment

Additional text

Inventory management tab**Recycle bin**

All material buffers that are declared as recycle bins are deleted or archived at regular intervals, depending on the configuration or customized settings. The process that deletes or archives the batches/lots included in material buffers is integrated in the scheduler. You can use the data management to configure the period after which batches/lots are deleted or archived.

Batches/lots that are located in a material buffer specified as a *recycling bin* cannot be logged on anymore. In this case, you have to manually repost them to a different material buffer in advance.

Batches cannot be logged on as input batches to a machine if the upstream input buffers of the machine are identified as a recycle bin.

Include in stock

Use this option to define whether the batches/lots located in this material buffer should be integrated in the stock overview.

Retention period

Duration of the storage period in the buffer (currently not used).

Hierarchy tab**Hierarchy**

Specifies the hierarchy level that is assigned to the buffer.

Hierarchical buffer

Specifies the assignment to the higher-level hierarchical buffer. A buffer can only be assigned to a hierarchical buffer with a hierarchy (number) that is greater than that of the assigned buffer.

Batch transport tab

Type

- No buffer
Refers to one of the below-mentioned buffer versions.
- Input buffer, output buffer
To configure the factory-wide batch transport, you can specify in this field whether the material buffer is an input or output buffer. A corresponding system is defined for an output buffer. This system specifies where the data posted to this material buffer is transported.



If you configure a material buffer as “input buffer”, the transport status of the batches/lots posted to this buffer will be set to “I” = initial. Consequently, these batches cannot be logged on to AIP.

Virtual storage buffer

You can use this field to define a material buffer as stock posting buffer. In this way, you can also specify whether the material buffer is a virtual storage buffer.

If you identify the material buffer as a placeholder for an external storage system, you can perform specific inventory queries for the virtual stock at a later point in time. This option is currently not used.

Corresponding system

If you configure a corresponding system, you can specify where all batches/lots included in this material buffer are transferred. For transport output buffers and stock posting buffers, you can enter the corresponding system where the batches/lots are transferred here.

Stock posting buffer

You can define a material buffer as a stock posting buffer to carry out stock postings. Batches are transferred by the standard batch interface using a file that is created in a directory of the file system. The corresponding system specifies this directory.

6 Inventory Limits

Summary

Menu	Master data → Material → Inventory limits
Transaction code	invl
Function authorization	invl

This function is used to create or modify inventory limits in the system.

Usage

This dialog is used to assign those materials to the [material buffers](#) that are temporarily stored in this [material buffer](#). The assignment enables the minimum stock to be specified in the buffer.

Integration

The values stored here are used in the following functions:

- Material management → [Stock monitoring](#)
- Operating facilities management → Graphic [machinery](#)
- Production control → [Graphic planning](#)

Requirement

[Material buffer](#) and units must be defined.

Selection criteria

The following selection criteria are available in the application:

Material buffer

Displays all inventory limits for the selected material buffer.

Material

Displays all inventory limits for the selected material.

When using multiple selection criteria, the amount of overlap of the selection criteria is displayed.

Field descriptions

Material

Unique material key

MBuffer

Name of the assigned material buffer

Unit

Quantity unit in which the inventory is listed

Minimum stock

Minimum stock of the material in the material buffer

Alert Min. stock level

Quantity of stock at which an alert occurs for increasing inventory

Min. warn. stock level

Quantity of stock at which a warning occurs for checking inventory

Max. warn. stock level

Quantity of stock at which a warning occurs for checking inventory

Alert Max. stock level

Quantity of stock at which an alert occurs for decreasing inventory

Max. stock level

Maximum material stock level in buffer

Comment

Additional comments

7 Transport Units

Summary

Menu	Material → Master data → Transport units
Transaction code	tu
Function authorization	tu

Usage

Use this function to create or to change transport units in the system.

Transport units are defined as those load carriers that are used to transport materials. When output batches are recorded at the terminal, the assignment to the producing [material type](#) can serve as selection that is saved to the batches.

Integration

The transport units will be shown in the evaluations in material management.

Selection criteria

The following selection criteria are available in the application:

Transport unit

Only the selected transport unit will be used.

Field descriptions

Transport unit

Unique identification of a transport unit

Designation

Clear text description of the transport unit

Quantity

Quantity of available transport units

Width

Width of the transport unit in the assigned unit. Currently used only for terminal display features.

Height

Height of the transport unit in the assigned unit. Currently used only for terminal display features.

Length

Length of the transport unit in the assigned unit. Currently used only for terminal display features.

Weight

Weight of the transport unit in the assigned unit. Currently used only for terminal display features.

Inventory management

Indicator relating to the inventory management of the transport unit. If this indicator is not set, this transport unit will also not be taken into account in the evaluation of the stock overview transport units

(Menu: *Material management* > *Inventory management* > *TPU stock overview*).

8 1 Assignment of Transport Unit to Material Type

Summary

Menu	Master data → Material → TPE assignment - Material type
Transaction code	astrum
Function authorization	astrum

Usage

This function is used to assign the corresponding [transport units](#) to a [material type](#) in the system.

Integration

For each [material type](#), several possible [transport units](#) can be defined. Then, when forming an output batch, the corresponding [transport unit](#) can be selected from those for the [material type](#) of the material to be produced.

Requirement

The transport units must be defined in the system.

Editing functions

Copy

There is the functionality to copy assignments from one material type to another. It can be chosen to either copy all assignments or only the missing ones.

Selection criteria

The following selection criteria are available in the application:

Material type

Only materials with the selected [material type](#) are selected.

Transport unit

Only materials with the selected [transport unit](#) are selected.

When using multiple selection criteria - if nothing else is specified - the amount of overlap of the selection criteria is displayed.

Field descriptions

Material type

Designation of the assigned [transport unit](#).

Transport unit

Designation of the assigned [transport unit](#).

Standard

The preferred [transport unit](#) for transporting materials of the specified material type.

Quantity

Quantity of the materials of this [material type](#) that match with this [transport unit](#). Used in the evaluation overview of [transport units](#) available.

Unit

Not in current use.

9 Batch data overview

Overview

Menu	Material management → Inventory management → Batch data overview
Transaction code	batov
Function authorization	batov

Available user fields		
Where	Object type/user field key	Source (type)
Detail view	CNR/SYSTEM	Batch (MF-D)
How to configure user fields? Which user field types are available?		

Purpose

Use the *batch data overview* to view or edit one or more batches depending on the entered selection criteria.

Integration

The *batch data overview* shows all existing batches matching the entered selection criteria.

The *batch data overview* shows the selected batches and their current status including further details.

Selection criteria

The application provides the following selection criteria:

"Batch" category

Batch number

The batch number represents the batch ID used by the user. The user can:

- enter/view the batch number in terminal dialogs
- use the batch number for tracing or information search via the office client



If the configuration option "automatic generation of the batch number" is enabled, leave the "batch number" field empty in the editing dialog when you add a new batch. The system will only assign the batch number automatically, if the "batch number" field is empty.

Material

Enter the material number as a selection criterion to display all batches assigned to this material number.

Workplace

Enter the workplace as a selection criterion to display all batches produced at this workplace/machine.

MES order number

MES order number (order/operation number) that produced the batch.

Internal batch number

The batch number is a **unique, system-wide** batch identification number.

Material buffers

The individual batches are located in a specific [Material buffers](#). Enter the material buffer to view all batches assigned to the selected material buffer.

Material type

The individual batches or materials belong to a [Material type](#). The same transport and handling guidelines apply to these material types across the system. Enter the material type to view all batches assigned to the selected material type.

Material category (kind of material)

The material category assigns batches to specific classes/ groups. Enter the material category to view all batches assigned to the selected material category.

Material name

The system takes over the material name from the currently logged in order when output batches are generated at shop floor terminals. As no material master is managed, the material name is stored redundantly in the batch description.

Manufacturing date from / until

Date/period of the production of a batch.

Consider long-term data

You can also select archived data.

Historic batch entries

If you use the function *historic batch entries* and you also use throughput batch numbers or serial numbers, the application will also show batches with the status "processed" for every throughput batch number or serial number. Use the function *historic batch entries* to get further process information on a throughput batch number or serial number.

Status category**Batch class**

The batch class describes the overall quality of the batch. Enter the batch class to view all batches matching the selected batch class.

Batch status

The batch status describes the technical system and production status of a batch. Enter the *batch status* as a selection criterion to view all batches assigned to this status.

Quality status

The quality status "blocked" prevents a batch from being logged on. Enter the quality status as a selection criterion to view all batches assigned to this quality status.

Manual Q status

Enter the manual Q status as a selection criterion to view all batches assigned to this status.

Material status

The material status indicates a logical status of the batch, e.g. packed, tested. Enter the *material status* as a selection criterion to view all batches assigned to this material status.

Transport status

The transport status represents the technical system status with respect to transfer postings to external storage. Select the transport status as a selection criterion to display all batches that are assigned to this transport status.

"Advance logon" option

Use this option to restrict the data to [batches logged on in advance](#).

"Attributes" category**Attribute (1 to 10)**

You can use the attributes directly stored with the batch to restrict the data displayed.

"Batch attributes" category**Batch attribute (name)**

You can use the [batch attributes](#) configured for the material type to restrict the data displayed. There are 40 text fields, 20 numeric fields and 20 decimal fields that may be configured.

"Alternative batch numbers" category**Alternative batch number (1 to 20)**

Select an alternative batch number as a selection criterion to display all batches that are assigned to this number.

"Reservation" category**Reserved for order**

Enter or select the order number in this field to display all batches that were produced for this order/ OP.

Reserved for OP

Enter or select the order/OP number in this field to display all batches that were produced for this order/ OP.

Dates category**Expiry date from / until**

Date/period that indicates the shelf life of a batch.

Availability date from / until

Date/period of the availability of a batch.

Warning time

Warning date of a batch

"Miscellaneous" category**Serial number**

Enter a *serial number* as a selection criterion to display all batches that are assigned to this serial number via the *serial number* field.

Batch number

Select the batch/lot number as a selection criterion to display all batches that are currently assigned to this number.

Person

Select *person* as a selection criterion to display all batches that have been produced by this person and, as a result, are currently assigned to this person (personnel number).

Collective batch (merged batch)

Enter a *collective batch number* (merged batch number) as a selection criterion to display all batches that are assigned to this collective batch number in the *collective batch* field (*merged batch* field).

PPS batch

Enter a *PPS batch* as a selection criterion to display all batches that are assigned to this PPS batch number in the *PPS batch* field.

Editing functions

Use the following functions that are available in addition to the standard features to edit one or several data records:



Add batch

Use this function to insert a new batch. Goods movements are not generated for the new batch.



Copy batch

Use this function to copy a batch. However, you have to enter a new batch number for the new batch. Goods movements are not generated for the new batch.



Edit batch

Use this function to edit a batch. But neither cancellations nor goods movements are generated for the changed batch.



Delete batch

Use this function to delete a batch. But neither cancellations nor goods movements are generated for the deleted batch.



Merge batches (new batch number)

Merging batches. A new batch number is generated for the new batch ([Merge batches](#)).

When merging collective batches, all individual batches are assigned to a new collective batch number.



Merge batches (use existing batch numbers)

Merging batches. A batch number already included in the merged batches is used for the new batch ([Merge batches](#)).

When merging collective batches, all individual batches are assigned to the selected collective batch number that is part of the merger.



Split batch

Create a new batch by splitting it off from an existing batch ([Batch split](#)). In this case, you can choose from the following options to deal with the remaining target quantity of the original batch:

- Repost the remaining quantity of the batch to the new batch
- Reduce the existing batch to the remaining quantity

If you split a collective batch, the selected individual batches/serial numbers are transferred to a batch split off. The remaining individual batches/serial numbers remain assigned to the original collective batch.



Repost

Use this function to repost a batch to another material buffer ([Repost batch](#)).



Generate

Use the function "enter goods receipt batch" to create batches manually ([Generate batch](#)). This is necessary if material is delivered via the incoming goods, for example.



Edit batch attribute

The tab "batch attributes" shows the [batch attributes](#) of the batch selected in the grid. The application only shows those batch attributes that have been assigned the option "show attribute on client" within the configuration of [batch attributes](#).

Provided that [batch attributes](#) are shown, click the button "edit batch attribute" to edit the batch attributes. In this case, the system does not check the field types/field lengths specified in the configuration of [batch attributes](#).

Go to



Graphic batch tracing

Calls up the application [Graphic batch tracing](#).



Document management

Calls up the [document management](#)

"Batch data overview" detail application

The *batch data overview* detail application shows all batches matching the entered selection criteria.

The documents entitled [batch object](#) and [batch structure](#) outline the information or field descriptions of the selected batches and, thus, of the detail application.

10 Generating Batch Numbers

Usage

The batch number is generated to basis 35; the digit specifications are relative and refer to the fixed proportion. The fixed proportion can be deduced from the prefixes for the automatically generated batch numbers defined in the [basic settings](#).

Digit 1-2	Terminal number
Digit 3	last digit of the year (starting at 1998)
Digit 4-5	Day of the year
Digit 6...	running number, distributed along the length of the batch number

The number generated at the terminals is unique for each terminal, i.e. there is no batch number 4711 for a goods receipt batch (create a batch) and a production batch (output batch change) on one day in a year.

11 Interrupting/Finishing OP without the Last Batch

Summary

The last batch for which a quantity may be entered when interrupting/logging the OP off is generated automatically between the last output batch change and logging the operation off.

But this system behavior is not always wanted. It might also be required that the last output batch is posted specifically by the user. Then the user directly interrupts/logs the OP off, whereas "no" last output batch is posted and/or no last output batch is generated for the operation.

Usage

The system immediately generates an output batch, once an OP is logged on. Even after an output batch change, a subsequent batch is generated immediately after completing the preceding output batch.

This cannot be changed in the system, as the system never knows in advance when the last output batch will be produced and/or if the next output batch will actually be the last one.

For this reason, it is ensured that:

- when interrupting/logging an operation off, the output batch generated at last will no longer be used and directly deleted.
- the last output batch is not visible on MOC and there is no connection to the running input batches of the operation.
- no last output batch is visible and, thus, no goods movement is generated.
- in posting records no quantities are written in the H record for a last batch.

Restrictions

- Cannot be used when batch numbers are assigned manually:
At the moment the function is only supported if batch numbers are assigned automatically at AIP. If the batch number is entered/scanned manually by the user, the function cannot be used, as the output batch entered at last must not be deleted before the OP is logged off/finished. If users can no longer enter "numbers" when changing the output batch, they cannot perform the output batch change and they have to interrupt/log the OP off.

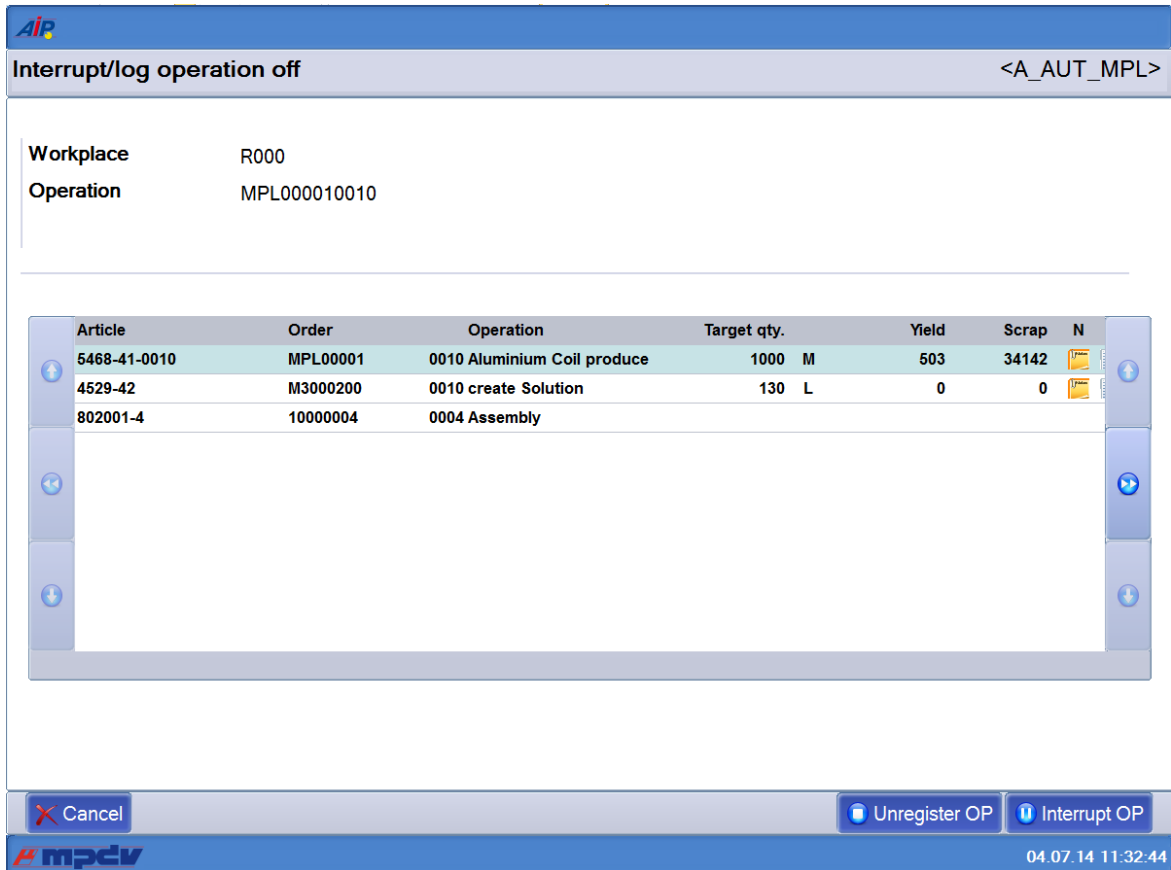
Procedure

The procedure to enter data "without" the last output batch when interrupting/logging off the OP is described as follows:

- Users log the operation on.

- Users register the input batches.
- Users perform the output batch change, once they want to finish an output batch.
- Once users have performed the last output batch change, they interrupt and/or log the operation off by using the function "interrupt/log OP off".

A running OP may be interrupted or logged off by clicking the "interrupt/log OP off" option. Then a dialog opens, where it may be chosen between "interrupt OP" and "log OP off".



Article	Order	Operation	Target qty.		Yield	Scrap	N
5468-41-0010	MPL00001	0010 Aluminium Coil produce	1000 M		503	34142	
4529-42	M3000200	0010 create Solution	130 L		0	0	
802001-4	10000004	0004 Assembly					

If "log OP off" or "interrupt OP" is clicked the logoff dialog opens containing the same input fields like the "output batch change" dialog.

If it was configured in the system that "no" last output batch is to be used when interrupting/logging the OP off, no output batch will be displayed, as there is no visible, active output batch at this point in time.

Unregister OP				<A_AB_MPL>	
Workplace	R000	Material	5468-41-0010		
Operation	MPL000010010	Batch	PR4RH69116		
Target buffer	MP_N_50505				
Transp. unit	ROLLCONT				
Comment					
Quality	Qty		Interrup. reason		
☒ Yield		0 M		1	
☐ Scrap	Reason	0	Deviation Reason	0	
Staff Number	☐ Memorize				
Cancel		Change i.batch		Preced. batches	Unregister OP

Configuration

These configurations have to be made in the system to enable the function.

Result

Data collection:

These results apply for data collection:

- No output batch is displayed/used in the dialog A_UN_MPL/ A_AB_MPL.
- No last quantity can be entered
- Additional information (e.g. material buffer, TPU, etc.) cannot be entered

Processing:

These results apply for processing:

- Provided that the relevant flag is set for the material type, the output batch generated by the interruption/logoff is directly assigned the status "D" that stands for deleted. The next archiving session considers this batch and deletes it.
- The "deleted" batch is not shown within tracing (graphic/tabular batch tracing).

- The deletion event is not displayed for the "deleted" batch in the batch history.
- A last H record is generated including the durations recorded between the last output batch change and interrupting/logging the OP off. The H record only includes durations and no quantities.
- How to proceed with automatic quantities:
If automatic quantities are recorded between the last output batch and logging the OP off, they are currently not taken into account.

12 Batch Consumption

Usage

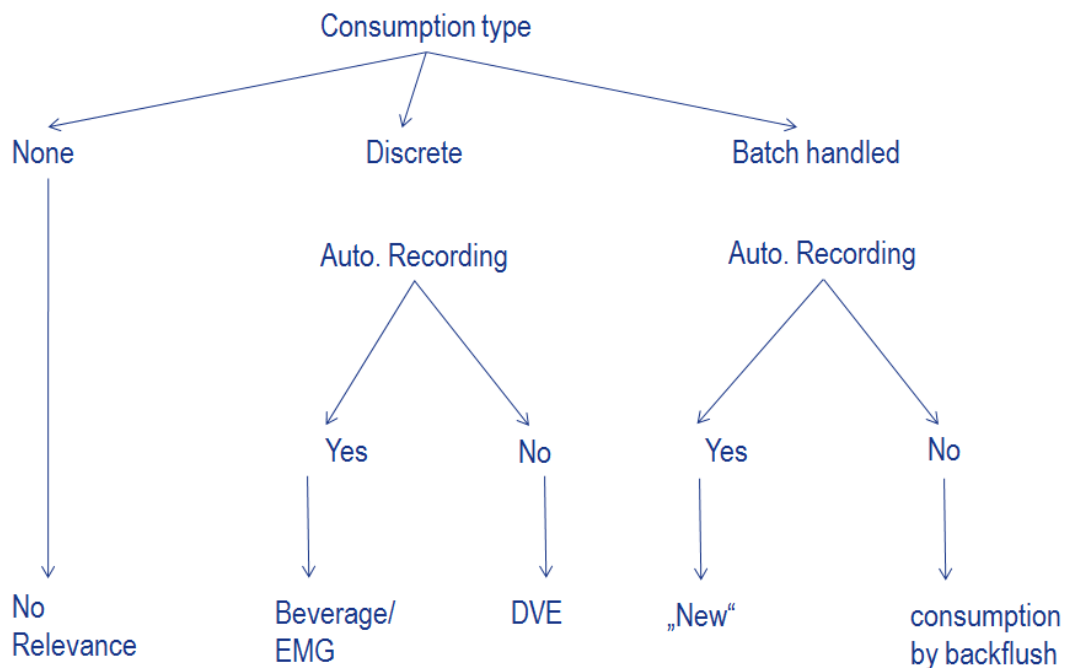
Material is used and represented in the system by:

- material postings to regulate inventories with/without ERP and without tracing
- batch-related material postings to trace back the recorded parts/materials

Subject to the type in use, consumption can be recorded differently in the system.

Procedure

These types of consumption recording are used in MES:



Discrete consumption recording:

Discrete consumption recording is used every time when a discrete amount of consumption can be entered for the used components by the user or a counter. Data is only entered for the component/material number. Data may be collected:

- automatically (configuration of a consumption meter per material)
- manually (the user enters material consumption manually)

Batch-related consumption recording:

Batch-related consumption recording is used every time when batches are used for the components for which a consumed quantity can be entered by the user or a meter. Data is collected regarding the input batch. Data may be collected:

- manually (dialog to log off the input batch and to enter batch consumption manually)
- in a retrograde manner/backflush (automatic calculation of batch consumption by generating the output batch quantity)
- automatically (automatic collection of batch consumption by a consumption meter)

Discrete consumption - manual



How to enter discrete consumption is described [here](#).

Discrete consumption - automatic

General

Automatically recorded consumption is collected by a meter configured at the machine. Data is collected for a material type of materials included in the component list.

Configuration

These configurations have to be set in the system if material consumed discretely is to be indicated/counted by a meter:

- Component of the OP:
The "consumption type" has to be set to "D = discrete".
- Material type of the material:
The option "inventory management" has to be set to "N = No".
- Meter for the material type of the material:
Configure meter like MDE meters.
Option "compensation with material" = yes
Option "material type" = material type of the material that is consumed

Posting/result

A goods issue is generated for automatically recorded consumption in the system.

Manual batch consumption

General

Manual batch consumption is entered by the input batch change function. The user enters the consumed quantity when logging the used input batch off.

Configuration

These configurations have to be set in the system if material is to be consumed manually as input batch:

- Component of the OP:
The option "consumption type" has to be set to "L = Backflush/with batch reference (retrograde)".
- Material type of the material:
The option "inventory management" has to be set to "E = Yes, when logging input batch off".

Posting/result

The consumed quantity is deducted from the remaining quantity of the input batch and the batch shows the reduced "remaining quantity" and the initial quantity.

A goods issue is generated for consumption in the system.



How to enter batch-related consumption is described here.

Retrograde batch consumption

General

Retrograde batch consumption is calculated continuously as the output batch quantity increases. When logging the input batch off, the remaining quantity of the input batch is reduced by the calculated consumption quantity. Usually, the user does no longer enter a quantity when logging the used input batch off.

Configuration

These configurations have to be set in the system if material is to be consumed in a retrograde manner as input batch:

- Component of the OP:
The option "consumption type" has to be set to "L = Backflush/with batch reference (retrograde)".
- Material type of the material:
The option "inventory management" has to be set to "R = Yes, backflush (retrograde)" or "G = Yes, backflush (only with YIELD batch), retrograde".

Posting/result

The quantity calculated in a retrograde manner is deducted from the remaining quantity of the input batch and then the batch shows the reduced "remaining quantity" and the initial quantity.

A goods issue is generated for consumption in the system.

Automatic batch consumption

General

The automatically recorded batch consumption is collected continuously as the meter quantity increases. When logging the input batch off, the remaining quantity of the input batch is reduced by the automatically recorded consumption quantity. Usually, the user does no longer enter a quantity when logging the used input batch off.

Configuration

- Component of the OP:
The option "consumption type" has to be set to "L = Backflush/with batch reference (retrograde)".
- Material type of the material:
The option "inventory management" has to be set to "R = yes, backflush (retrograde)".
- Meter for the material type/BOM item of the affected material:
Configure meter like MDE meters.
Option "compensation with material" = yes
Option "material type" = material type of the material that is consumed
or
Option BOM item = BOM item of the material that is consumed



If the BOM item is used within meter configuration, it is important that within the OP's component list the material is always used as the same BOM item (from ERP work plan).

Posting/result

The automatically recorded quantity is deducted from the remaining quantity of the input batch and then the batch shows the reduced "remaining quantity" and the initial quantity.

A goods issue is generated for consumption in the system.

13 MPL for the coil-based production

Purpose

You use this documentation, if one or several of the following conditions are true for you.

- You use the product "MPL" for the processing of rolls (coils) in your production and you want to be informed about the configuration options.
- You want to use the product "MPL" for the processing of rolls in your production.
- You want to be informed about the possibilities of specific functions that are designed for a production using rolls.



The functions that are designed for a coil-based production are included in the product group "MPL".

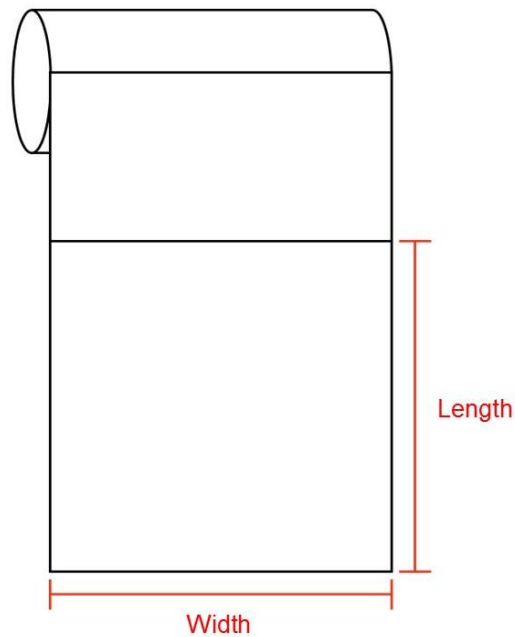
The functions for the coil-based production do not form a separate product. They are part of the product "MPL".

General information

You use the office client to edit the required values for a production process using rolls. You configure the respective values of the operation to this end. Go to the application "Edit operations", tab "CBM". The fields of tab "CBM" of an operation are described in the documentation [MES-Operation structure](#).

Using the terms "Length" and "Width"

For the coil-based manufacturing, HYDRA has defined the use of the terms "length" and "width". The following graphic demonstrates what is called "length" and what is called "width" in HYDRA.



Using the unit "Surface per piece"

HYDRA specifies the surface per piece in the unit mm^2/pce (square millimeters per piece). The specified surface per piece refers to the production of one article.

Mother/child operation

You can use the mother-child relation of operations for a production using rolls. The mother-child relation defines the relation of one mother operation to one or more child operations. If you define a mother-child relation, the child operations are automatically logged on when you log on the respective mother operation.

The sequencing list of the shop floor client only displays the mother operations (option "Branch OP" = "M"). The sequencing list does not show child operations (option "Branch OP" = "K").

Other terms

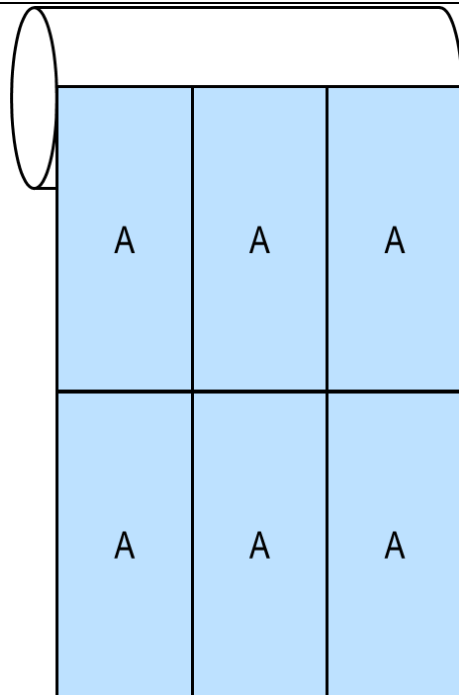
Seam width

If you cut material that is stocked on rolls, you sometimes cut off the edges. In HYDRA, the width of this cut off edge is called seam width.

Cutting plans

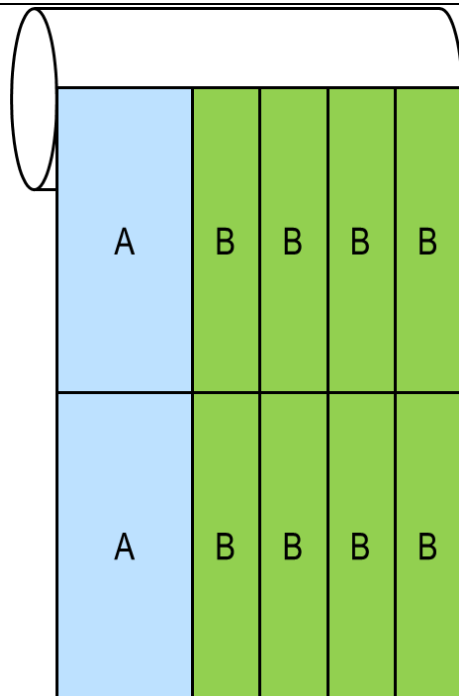
There are two types of cutting plans: homogeneous and heterogeneous (chaotic) cutting plans.

Homogeneous cutting plan: exactly one article



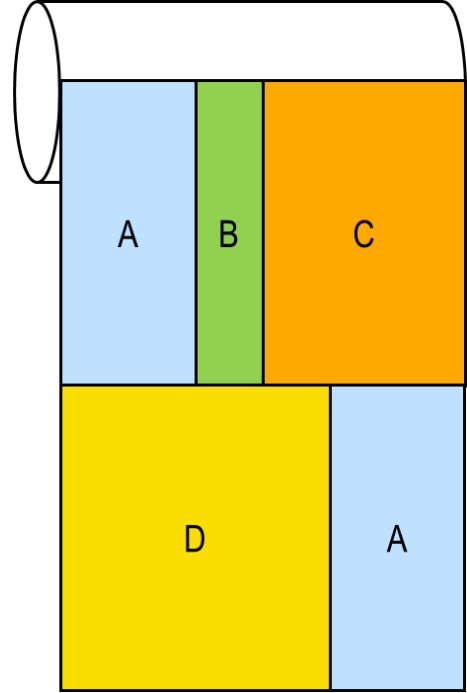
- For an order, several daughter rolls with identical width are produced. The order includes exactly one cutting process.
- A mother roll is cut in n rolls of identical width.
- A cut across the material width results in several daughter rolls of identical length.
- The individual cuts across the material width can result in different lengths. Usually, only the last part (after the last cut) has a different length.

Homogeneous cutting plan: several articles



- An order produces different output widths. The order includes different cutting processes in the mother-child relation.
- The mother roll is cut into n daughter rolls.
- A cut across the material width results in several daughter rolls of identical length.
- The individual cuts across the material width can result in different lengths. Usually, only the last part (after the last cut) has a different length.

Heterogeneous (chaotic) cutting plan: several articles

	• An order produces different output widths. The order includes different cutting processes in the mother-child relation. • The mother roll is cut into n daughter rolls. • A cut across the material width results in several daughter rolls of identical length. • The individual cuts across the material width can result in different lengths. Usually, only the last part (after the last cut) has a different length. • With this version, the cutting plan of the order is not uniform and can be different for each cut across in an extreme case.
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Further notes/restrictions



For workplace type "S" (cutting unit), the run-through batch processing is not supported.

Configuration

The configuration is described in the document [Configuration of coil-based manufacturing](#).